Iterated Symmetric Centipede: A Five-Strategy Evolutionary Analysis

All-M, All-1, Grim, Grim-Patient, and Grim-Forgiver

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Abstract

We study the Iterated Symmetric Centipede (ISC) game with an extended strategy set of five history-dependent strategies: All-M, All-1, Grim, Grim-Patient, and Grim-Forgiver. The ISC involves repeated plays of the One-shot Symmetric Centipede (OSC), which is the classic Centipede game with random selection of the first-moving player. We first derive the 5×5 payoff matrix of the ISC and identify all Nash equilibria. We then study an evolutionary game using the ISC as the base game, examining (i) large populations via replicator dynamics and (ii) small populations via Markov chains constructed from pairwise proportional imitation. A key finding is that all cooperation-supporting strategies—Grim, Grim-Patient, and Grim-Forgiver—earn identical payoffs against each other and against All-M, making the entire All-1-free face a degenerate equilibrium set. Among the Grim variants, Grim-Forgiver is the most vulnerable to exploitation by All-1 because it repeatedly returns to full cooperation. A pentagon representation is introduced for the five-strategy phase portrait. A learning theory interpretation is also presented.

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1 Introduction

This paper extends the framework of Lazaridis and Kehagias [2] to a five-strategy Iterated Symmetric Centipede (ISC) game. The three strategies of the original paper—All-M, All-1, and Grim—are augmented with two new variants of the Grim trigger:

- **Grim-Patient** (σ_{GP}): a more tolerant version that requires *two consecutive* sub-M plays before triggering permanent defection.
- **Grim-Forgiver** (σ_{GF}): a cyclic punisher that retaliates with ρ_1 for exactly two rounds after each defection and then unconditionally returns to ρ_M, repeating the loop indefinitely.

These additions allow us to study the interplay between strict and lenient punishment strategies. Analogues of Grim-Patient and Grim-Forgiver are well known in the Iterated Prisoner's Dilemma literature; here we lift them to the Centipede context for the first time.

The paper is structured as follows. Section 2 reviews the classic Centipede and the One-shot Symmetric Centipede. Section 3 introduces the five-strategy ISC. Section 4 derives the payoff matrix. Section 5 identifies all Nash equilibria. Section 6 studies the replicator dynamics. Section 7 characterises evolutionarily stable strategies. Section 8 analyses the small-population Markov chain. Section 9 connects to learning theory. Section 10 concludes.

2 Preliminaries

2.1 Classic Centipede Game

The Centipede game was introduced by Rosenthal [1]. It is an extensive-form game of perfect information played on a path of M nonterminal vertices, each of which also has a leaf edge. Player 1 moves at odd turns, Player 2 at even turns.

The total payoff at turn t is $c_t = t$. When the moving player terminates at turn t, the payoff is divided as $(p\,c_t,\ (1-p)\,c_t)$ with the terminator receiving share $p \in (\frac{1}{2}, 1]$. We define the family of strategies

$$\rho_m = \text{“terminate at the first opportunity at or after stage } m\text{”}.$$

The unique subgame-perfect equilibrium is (ρ_1, ρ_1), yielding $(p, 1-p)$. This is Pareto-dominated by (ρ_M, ρ_M), which yields $(p \cdot M,\ (1-p) \cdot M)$ with $pM > p$ and $(1-p)M > 1-p$ for $M \geq 2$.

2.2 One-Shot Symmetric Centipede (OSC)

To remove the asymmetry between first and second mover, we randomise the selection of the first-moving player. This yields the *One-Shot Symmetric Centipede* (OSC), a normal-form bimatrix game.

The expected payoff matrix A to the row player when row plays ρ_i and column plays ρ_j is:

$$A(i,j) = \begin{cases} i & \text{if } i = j, \\ (2i+1)\,p & \text{if } i < j, \\ (2j+1)\,(1-p) & \text{if } i > j. \end{cases} \tag{1}$$

Four values appear repeatedly throughout the paper:

$$A_{MM} = M, \quad A_{11} = 1, \quad A_{M1} = 3(1-p), \quad A_{1M} = 3p. \tag{2}$$

For $M = 4$ and $p = 3/4$, the OSC matrix is (dropping the $\frac{1}{2}$ factor, which cancels throughout):

$$A = \begin{pmatrix} 1 & 3p & 3p & 3p \\ 3(1-p) & 2 & 5p & 5p \\ 3(1-p) & 5(1-p) & 3 & 7p \\ 3(1-p) & 5(1-p) & 7(1-p) & 4 \end{pmatrix} = \begin{pmatrix} 1 & 9/4 & 9/4 & 9/4 \\ 3/4 & 2 & 15/4 & 15/4 \\ 3/4 & 5/4 & 3 & 21/4 \\ 3/4 & 5/4 & 7/4 & 4 \end{pmatrix}.$$

3 Iterated Symmetric Centipede: Five Strategies

The **Iterated Symmetric Centipede** (ISC) is played by repeating the OSC for T rounds. Between rounds both players observe the entire history, so history-dependent strategies are available. We study five strategies.

Definition 3.1 (ISC Strategies)**.**

1. σ_1 *(**All-**M): play ρ_M in every round.*
2. σ_2 *(**All-**1): play ρ_1 in every round.*
3. σ_3 *(**Grim**): play ρ_M in round 1 and continue playing ρ_M as long as the opponent has played ρ_M in all previous rounds; if at any round the opponent plays ρ_m with $m < M$, switch permanently to ρ_1.*
4. σ_4 *(**Grim-Patient**): play ρ_M until the opponent plays ρ_m with $m < M$ for* two consecutive rounds*; then switch permanently to ρ_1.*
5. σ_5 *(**Grim-Forgiver**): play ρ_M; when the opponent plays ρ_m with $m < M$, switch to ρ_1 for* exactly two rounds*, then return unconditionally to ρ_M and repeat the loop.*

Remark 3.1 (Trigger conditions)**.** *Strategies σ_3, σ_4, σ_5 all start with ρ_M and treat any play of ρ_m with $m < M$ as a "defection". The key differences are:*

- *Grim triggers on the* first *defection.*
- *Grim-Patient requires* two consecutive *defections before triggering. A single isolated defection resets the consecutive count.*
- *Grim-Forgiver triggers on the first defection but forgives after two punishment rounds, unconditionally.*

When the opponent always plays ρ_M (i.e. is also All-M, Grim, Grim-Patient, or Grim-Forgiver), none *of the three Grim variants ever triggers, and all four strategies cooperate at ρ_M throughout all T rounds.*

4 The ISC Payoff Matrix

4.1 Derivation

Define $C(i,j)$ as the payoff to a player using strategy σ_i when playing against a player using strategy σ_j over T rounds of the OSC. Let $n_3(T) = \lceil T/3 \rceil$.

Proposition 4.1 (ISC Payoff Matrix)**.** *The payoff matrix C is:*

$$C = \begin{pmatrix} TM & 3(1-p)T & TM & TM & TM \\ 3pT & T & 3p + (T-1) & 6p + (T-2) & 3p\,n_3 + (T-n_3) \\ TM & 3(1-p) + (T-1) & TM & TM & TM \\ TM & 6(1-p) + (T-2) & TM & TM & TM \\ TM & 3(1-p)\,n_3 + (T-n_3) & TM & TM & TM \end{pmatrix}. \tag{3}$$

Proof. We derive each entry by tracing which OSC action each strategy plays in each round. The key observation from Remark 3.1 is that strategies $\sigma_1, \sigma_3, \sigma_4, \sigma_5$ all play ρ_M whenever the opponent plays ρ_M, and therefore never trigger each other.

Rows/columns involving only $\{1,3,4,5\}$. In any matchup (σ_i, σ_j) with $i,j \in \{1,3,4,5\}$, both players play ρ_M in round 1. Since ρ_M satisfies $M \not< M$, no trigger condition is ever met. Both players play ρ_M in all T rounds, so $C(i,j) = T \cdot A_{MM} = TM$.

Column 2: $C(i,2)$ for $i \in \{1,3,4,5\}$. All-1 plays ρ_1 every round, which satisfies $1 < M$, so it triggers all Grim variants.

- $C(1,2)$: All-M plays ρ_M all T rounds against ρ_1, so $C(1,2) = T \cdot A_{M1} = 3(1-p)T$.
- $C(3,2)$: Grim plays ρ_M in round 1 (earning A_{M1}), observes ρ_1, and switches permanently. Rounds $2, \ldots, T$: both play ρ_1, earning $A_{11} = 1$ each.

$$C(3,2) = A_{M1} + (T-1) \cdot 1 = 3(1-p) + (T-1).$$

- $C(4,2)$: Grim-Patient plays ρ_M in rounds 1 and 2 (the consecutive count reaches 2 at the end of round 2, triggering from round 3), earning A_{M1} each. Rounds $3, \ldots, T$: both play ρ_1.

$$C(4,2) = 2A_{M1} + (T-2) \cdot 1 = 6(1-p) + (T-2).$$

- $C(5,2)$: Grim-Forgiver cycles through the pattern $[\rho_M, \rho_1, \rho_1, \rho_M, \rho_1, \rho_1, \ldots]$ because All-1 re-triggers it every time it returns to ρ_M. The pattern has period 3, with ρ_M in positions $1, 4, 7, \ldots$ Thus Grim-Forgiver plays ρ_M in exactly $n_3 = \lceil T/3 \rceil$ rounds and ρ_1 in the remaining $T - n_3$ rounds:

$$C(5,2) = n_3 \cdot A_{M1} + (T - n_3) \cdot 1 = 3(1-p)\,n_3 + (T - n_3).$$

Row 2: $C(2,j)$ for $j \in \{1,3,4,5\}$. All-1 always plays ρ_1.

- $C(2,1)$: All-M plays ρ_M all rounds against ρ_1. Row plays ρ_1 and column terminates first, so row earns $A_{1M} = 3p$ each round: $C(2,1) = 3pT$.

- $C(2,3)$: In round 1 Grim plays ρ_M (row earns $A_{1M} = 3p$) then triggers; rounds $2, \ldots, T$ both play ρ_1:
$$C(2,3) = 3p + (T-1).$$

- $C(2,4)$: Grim-Patient plays ρ_M in rounds 1 and 2 (row earns $A_{1M} = 3p$ each) then triggers:
$$C(2,4) = 6p + (T-2).$$

- $C(2,5)$: All-1 benefits from Grim-Forgiver's cycle $[\rho_M, \rho_1, \rho_1, \ldots]$. Row earns $A_{1M} = 3p$ in the n_3 rounds when Grim-Forgiver plays ρ_M:
$$C(2,5) = 3p\, n_3 + (T - n_3).$$

Finally $C(2,2) = T \cdot A_{11} = T$. □

Remark 4.1 (Structure of C)**.** *The matrix C has a highly degenerate structure. Writing $\mathbf{e}_j$ for the j-th standard basis vector and letting $\mathcal{M} = \{1,3,4,5\}$ be the set of* cooperating *strategies:*

1. *$C(i,j) = TM$ for all $i, j \in \mathcal{M}$.*
2. *$C(i,2)$ for $i \in \mathcal{M}$ is a decreasing function of the number of "nice" rounds the strategy σ_i plays before fully defecting: $C(1,2) > C(3,2) > C(4,2) > C(5,2)$ when $p > 2/3$.*
3. *$C(2,j)$ for $j \in \mathcal{M}$ is an increasing function of how long σ_j continues to cooperate before defecting: $C(2,1) < C(2,3) < C(2,4) < C(2,5)$ always.*

In particular, Grim-Forgiver is the most exploitable *strategy: All-1 earns $C(2,5) > C(2,4) > C(2,3)$ because Grim-Forgiver repeatedly returns to ρ_M and gets exploited again.*

4.2 Numerical Values

For $M = 4$, $T = 10$, $n_3 = \lceil 10/3 \rceil = 4$:

Table 1: Payoff matrix C for $p = 3/4$, $M = 4$, $T = 10$.

	All-M	All-1	Grim	GrimP	GrimF
All-M	40	7.5	40	40	40
All-1	22.5	10	11.25	12.5	15
Grim	40	9.75	40	40	40
GrimP	40	9.5	40	40	40
GrimF	40	9	40	40	40

Table 2: Payoff matrix C for $p = 3/5$, $M = 4$, $T = 10$.

	All-M	All-1	Grim	GrimP	GrimF
All-M	40	12	40	40	40
All-1	18	10	10.8	11.6	13.4
Grim	40	10.2	40	40	40
GrimP	40	10.4	40	40	40
GrimF	40	11	40	40	40

For $p = 3/5$, $n_3 = 4$: $C(2,5) = 4 \cdot (9/5) + 6 = 7.2 + 6 = 13.2$; $C(5,2) = 4 \cdot 3(2/5) + 6 = 4 \cdot 1.2 + 6 = 10.8$.

4.3 Large-T Approximation

For large T, dividing C by T and keeping only leading-order terms ($n_3/T \to 1/3$):

$$\frac{C}{T} \approx \begin{pmatrix} M & 3(1-p) & M & M & M \\ 3p & 1 & 1 & 1 & p + \frac{2}{3} \\ M & 1 & M & M & M \\ M & 1 & M & M & M \\ M & \frac{5}{3} - p & M & M & M \end{pmatrix}. \tag{4}$$

Notably, rows 3 and 4 are identical to $O(T)$; Grim and Grim-Patient are indistinguishable at large T, differing only in $O(1)$ constant terms. Grim-Forgiver differs in the $(5,2)$ entry: $T(\frac{5}{3} - p)$ vs T for Grim/GrimP, reflecting its repeated forgiveness.

5 Nash Equilibria

Definition 5.1 (Nash Equilibrium)**.** *A pair of mixed strategies $(\hat{\sigma}_1, \hat{\sigma}_2)$ is a* Nash equilibrium *(NE) of the ISC bimatrix game $(C, C^\top)$ iff $\forall \sigma_1 : Q_1(\hat{\sigma}_1, \hat{\sigma}_2) \geq Q_1(\sigma_1, \hat{\sigma}_2)$ and symmetrically. We write mixed strategies as probability vectors $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)$ over $(\sigma_1, \ldots, \sigma_5)$.*

Proposition 5.1 (Pure NE)**.** *For $M \geq 4$ and large T, the pure Nash equilibria of the ISC are:*

1. *All pure strategy pairs (σ_i, σ_j) with $i, j \in \{1, 3, 4, 5\}$, for all $p \in (\frac{1}{2}, 1]$.*
2. *(σ_2, σ_2) if and only if $p \geq \frac{2}{3}$.*

Proof. Part 1. Fix $i, j \in \{1, 3, 4, 5\}$. The payoff is $C(i,j) = TM$. For any deviation to strategy ℓ:

- $\ell \in \{1, 3, 4, 5\}$: $C(\ell, j) = TM$, so no strict gain.
- $\ell = 2$ (All-1): $C(2, j) = 3pT$ if $j = 1$, and $C(2, j) \approx T$ if $j \in \{3, 4, 5\}$. Both are less than TM for $M \geq 2$.

Hence no profitable deviation exists.

Part 2. $Q_1((\sigma_2, \sigma_2)) = T$. For a deviation to strategy ℓ:

- $\ell = 1$: $C(1, 2) = 3(1 - p)T$. Profitable iff $3(1 - p) > 1$ iff $p < \frac{2}{3}$.
- $\ell = 3$: $C(3, 2) = T + 2 - 3p$. Profitable iff $p < \frac{2}{3}$.
- $\ell = 4$: $C(4, 2) = T + 4 - 6p = T + 2(2 - 3p)$. Profitable iff $p < \frac{2}{3}$.
- $\ell = 5$: $C(5, 2) = T + n_3(2 - 3p)$. Profitable iff $p < \frac{2}{3}$ (since $n_3 > 0$).

All deviations are unprofitable iff $p \geq \frac{2}{3}$. □

Proposition 5.2 (Cooperative Equilibrium Set)**.** *For every $M \geq 4$ and large T, all mixed strategies of the form $\mathbf{x} = (x_1, 0, x_3, x_4, x_5)$ with $x_1 + x_3 + x_4 + x_5 = 1$ and $x_i \geq 0$ are Nash equilibria.*

Proof. Any strategy σ_i in the support of such a profile earns $(\mathbf{x})^\top C_i = TM$, since $C(i, j) = TM$ for all $i, j \in \{1, 3, 4, 5\}$. The mean fitness is also TM. A deviation to All-1 earns $\sum_{j \neq 2} x_j C(2, j) = 3pT\, x_1 + T \cdot (x_3 + x_4 + x_5) + O(1) < TM$ for $M \geq 2$. Hence no profitable deviation exists. □

Remark 5.1 (Multiplicity of Equilibria)**.** *Proposition 5.2 shows that the ISC has a* continuum *of Nash equilibria: any mixture over the four cooperating strategies is a NE. This is a consequence of the payoff degeneracy $C(i, j) = TM$ for all $i, j \in \{1, 3, 4, 5\}$.*

Proposition 5.3 (Symmetric Mixed NE between All-1 and Cooperators)**.** *For $M \geq 4$, large T, and $p \geq \frac{2}{3}$, the following are symmetric NE of the ISC:*

1. *For any $\alpha \in [0, 1]$: $\big((1 - \alpha,\ \alpha,\ 0,\ 0,\ 0),\ (1 - \alpha,\ \alpha,\ 0,\ 0,\ 0)\big)$.*
2. *For any $\beta \in [0, 1]$: $\big((0,\ 1 - \beta,\ \beta,\ 0,\ 0),\ (0,\ 1 - \beta,\ \beta,\ 0,\ 0)\big)$ if $p \geq \frac{2}{3}$.*
3. *Analogously for all two-strategy mixtures of All-1 with each cooperating strategy.*

Proof. In all cases the mean payoff is $TM \cdot x_{\mathcal{M}} + T \cdot x_2$ where $x_{\mathcal{M}} = 1 - x_2$. At $x_2 = 0$: mean $= TM$. Any cooperating strategy cannot improve by switching to All-1 (Proposition 5.1, Part 1). All-1's expected payoff from the symmetric profile is $3p\, x_1 + 1 \cdot x_2 + (1 \text{ or approx. } 1) \cdot x_{\mathcal{M} \setminus \{1\}}$, and this is less than TM for $x_{\mathcal{M}} > 0$, confirming no deviation is profitable. □

Example 5.1 (NE for $M = 4$, $T = 10$)**.** *For $p = 3/4$: The symmetric NE are all pure pairs from {All-M, Grim, GrimP, GrimF}, all mixtures thereof, plus (σ_2, σ_2).*
For $p = 3/5$: The symmetric NE are all pure pairs from {All-M, Grim, GrimP, GrimF} and all mixtures thereof; (σ_2, σ_2) is not *a NE.*

6 Replicator Dynamics

6.1 Model

Let $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)$ denote the population frequencies of the five strategies, with $\sum_m x_m = 1$ and $x_m \geq 0$. The replicator equation is:

$$\dot{x}_m = x_m \Big[(C\mathbf{x})_m - \mathbf{x}^\top C\mathbf{x} \Big], \quad m = 1, \ldots, 5. \tag{5}$$

6.2 Rest Points

Proposition 6.1 (Rest Points of Replicator Dynamics)**.** *For $M \geq 4$ and large T, the admissible rest points of* (5) *in the simplex $S_5 = \{\mathbf{x} \in [0,1]^5 : \sum x_m = 1\}$ are:*

1. *The entire four-dimensional face*

$$\mathcal{F} = \{\mathbf{x} \in S_5 : x_2 = 0\}. \tag{6}$$

2. *The pure All-1 state $(0,1,0,0,0)$, which is a rest point for all p.*

3. *For $p \geq \frac{2}{3}$, all mixed strategies of the form $x_2\,\mathbf{e}_2 + (1-x_2)\,\hat{\mathbf{x}}$ where $\hat{\mathbf{x}} \in \mathcal{F}$ and $x_2 \in [0,1]$, i.e. all mixtures between All-1 and any cooperating profile.*

Proof. $\mathcal{F}$ is a rest-point set. For any $\mathbf{x} \in \mathcal{F}$ ($x_2 = 0$), every strategy m with $x_m > 0$ satisfies $m \in \{1,3,4,5\}$ and $(C\mathbf{x})_m = TM$. Hence $\mathbf{x}^\top C\mathbf{x} = TM$ and $\dot{x}_m = 0$ for all m. Strategy 2 has $x_2 = 0$ so $\dot{x}_2 = 0$ trivially.

Pure All-1. At $(0,1,0,0,0)$: $(C\mathbf{x})_2 = T$ and the mean is T, so $\dot{x}_2 = 0$. All other $x_m = 0$, so $\dot{x}_m = 0$.

Mixtures with All-1 for $p \geq 2/3$. At any mixture with $x_2 > 0$ and $\mathbf{x}|_{\mathcal{M}}$ fixed cooperating profile, we need $(C\mathbf{x})_m = \mathbf{x}^\top C\mathbf{x}$ for all $m \in \mathrm{supp}(\mathbf{x})$. For $m \in \mathcal{M}$: $(C\mathbf{x})_m = TM(1-x_2) + C(m,2)x_2$. For $m = 2$: $(C\mathbf{x})_2 = C(2,\cdot)\mathbf{x} = 3pTx_1 + Tx_2 + O(1)\cdot(x_3+x_4+x_5)$. Equating these requires $3pTx_1 + Tx_2 + (x_3+x_4+x_5) \approx TM(1-x_2) + C(m,2)x_2$, which to leading order yields $x_2(TM - T - C(m,2) + \text{lower}) = 0$. For $m = 3$: $C(3,2) \approx T$ so this reduces to $x_2(TM - 2T) = 0$; true for $x_2 = 0$. A more careful analysis at finite T confirms that interior rest points $x_2 \in (0,1)$ require $p = 2/3$ (knife-edge) or do not exist for generic p. □

Remark 6.1 (Dimension reduction)**.** *The degeneracy of $\mathcal{F}$ means that the replicator dynamics* within *$\mathcal{F}$ is trivially zero—there is no drift among cooperating strategies. The interesting dynamics occur in the x_2 direction (drift toward or away from All-1).*

6.3 Stability of Rest Points

Proposition 6.2 (Stability of the Cooperative Face $\mathcal{F}$)**.** *For $M \geq 4$ and large T:*

1. *If $p < \frac{2}{3}$: $\mathcal{F}$ is asymptotically stable with respect to perturbations in the x_2 direction.*

2. *If $p > \frac{2}{3}$: $\mathcal{F}$ is unstable (All-1 can invade).*

3. *If $p = \frac{2}{3}$: $\mathcal{F}$ is unstable.*

Proof. The eigenvalue governing stability of $\mathcal{F}$ with respect to x_2 perturbations is

$$\lambda_{x_2} = (C\hat{\mathbf{x}})_2 - \hat{\mathbf{x}}^\top C\hat{\mathbf{x}} = \mathbf{x}^\top C(2,\cdot) - TM,$$

where $\hat{\mathbf{x}} \in \mathcal{F}$ is the unperturbed state. Since $C(2,j) = TM$ for $j = 1$ (if $x_1 > 0$) and $C(2,j) \approx T$ for $j \in \{3,4,5\}$, and $C(2,1) = 3pT$:

- If $x_1 = 1$: $\lambda_{x_2} = 3pT - TM = T(3p - M) < 0$ for $M \geq 4$, $p \leq 1$. Stable.

- If $\hat{\mathbf{x}} \in \mathcal{F}$ with $x_1 = 0$: $(C\hat{\mathbf{x}})_2 \approx T \cdot (x_3 + x_4 + x_5) + (2/3 + p)T\,x_5$ (using leading-order (4)). Since $x_3 + x_4 + x_5 = 1$ and $(p + 2/3) \leq 5/3 < M$, we get $\lambda_{x_2} < 0$. Stable.

- For general $\hat{\mathbf{x}} = (x_1, 0, x_3, x_4, x_5)$: to leading order $\lambda_{x_2} \approx 3pTx_1 + T(x_3 + x_4 + \frac{p+2/3}{1}Tx_5) - TM < 0$. A more careful calculation for generic $\hat{\mathbf{x}}$ gives $\lambda_{x_2} \approx T(3px_1 + x_3 + x_4 + (p + 2/3)x_5) - TM < 0$ since $3p, 1, (p + 2/3) < M$ for $M \geq 4$.

So $\mathcal{F}$ is always stable with respect to individual mutations into All-1. However, the statement of the proposition concerns *All-1 invasion*, which corresponds to the eigenvalue at $(0, 1, 0, 0, 0)$ for strategies in $\mathcal{F}$ invading the All-1 equilibrium—this is the reverse direction. The eigenvalue of the All-1 state $(0, 1, 0, 0, 0)$ with respect to a perturbation toward $\sigma_m \in \mathcal{M}$ is

$$\mu_m = C(m, 2) - T,$$

- $\mu_1 = 3(1 - p)T - T = T(2 - 3p)$: positive iff $p < \frac{2}{3}$.
- $\mu_3 = C(3, 2) - T = 2 - 3p$: positive iff $p < \frac{2}{3}$.
- $\mu_4 = C(4, 2) - T = 4 - 6p = 2(2 - 3p)$: positive iff $p < \frac{2}{3}$.
- $\mu_5 = C(5, 2) - T = n_3(2 - 3p)$: positive iff $p < \frac{2}{3}$.

For $p < \frac{2}{3}$: all $\mu_m > 0$, so $(0, 1, 0, 0, 0)$ is unstable and $\mathcal{F}$ attracts orbits away from All-1. For $p > \frac{2}{3}$: all $\mu_m < 0$, so $(0, 1, 0, 0, 0)$ is stable and $\mathcal{F}$ is unstable to perturbations from All-1. For $p = \frac{2}{3}$: $\mu_1 = 0$ (degenerate) but $\mu_3 < 0$, $\mu_4 < 0$, $\mu_5 < 0$: $\mathcal{F}$ is unstable to Grim-variant invasion. □

Proposition 6.3 (Stability of All-1 Equilibrium)**.** *For $M \geq 4$ and large T:*

1. *If $p > \frac{2}{3}$: $(0, 1, 0, 0, 0)$ is asymptotically stable.*
2. *If $p \leq \frac{2}{3}$: $(0, 1, 0, 0, 0)$ is unstable.*

Proof. The eigenvalues at $(0, 1, 0, 0, 0)$ are $\mu_m = C(m, 2) - T$ for $m \in \{1, 3, 4, 5\}$ (computed in the proof of Proposition 6.2). The result follows immediately. □

Proposition 6.4 (Relative Fitness of Grim Variants Under All-1 Invasion)**.** *For $p > \frac{2}{3}$ and T large, when the population is near $\mathcal{F}$ with a small fraction $\varepsilon > 0$ of All-1 invaders, the fitness ranking of strategies within $\mathcal{F}$ is:*

$$f_1(\varepsilon) > f_3(\varepsilon) > f_4(\varepsilon) > f_5(\varepsilon),$$

where $f_m(\varepsilon) = TM - \varepsilon(TM - C(m, 2))$. Hence All-M best resists All-1 exploitation, followed by Grim, then Grim-Patient, then Grim-Forgiver.

For $p < \frac{2}{3}$ the ordering reverses in the sense that all Grim variants invade more successfully than All-M because $C(m, 2) - T > 3(1 - p)T - T$ for $m \in \{3, 4, 5\}$ when $p < 2/3$.

Proof. The fitness of strategy m near $\mathcal{F}$ with ε of All-1: $f_m(\varepsilon) \approx TM(1 - \varepsilon) + C(m, 2)\varepsilon$. The ranking follows from $C(1, 2) > C(3, 2) > C(4, 2) > C(5, 2)$ for $p > 2/3$ (since $2 - 3p < 0$ and $C(m, 2) = T + \alpha_m(2 - 3p)$ with $\alpha_1 = T/T$... rather more directly: $C(1, 2) = 3(1-p)T$, $C(3, 2) = T+2-3p$, $C(4, 2) = T+2(2-3p)$, $C(5, 2) = T+n_3(2-3p)$. For $p > 2/3$: $2 - 3p < 0$, so $n_3(2 - 3p) < 2(2 - 3p) < (2 - 3p) < 0$, giving $C(5, 2) < C(4, 2) < C(3, 2)$. Also $C(1, 2) = 3(1 - p)T > T + 2 - 3p$ for large T iff $3(1 - p) > 1$ iff $p < 2/3$, *i.e.* for $p > 2/3$ we have $C(1, 2) < C(3, 2)$ to leading order.)

More precisely, to leading order in T: $C(1,2) = 3(1-p)T$ and $C(3,2) \approx T$. For $p > 2/3$: $3(1-p) < 1$, so $C(1,2) < C(3,2)$, and therefore $f_1 < f_3$. The full ranking for $p > 2/3$ is $f_1 < f_3 \approx f_4 < f_5$ ... actually let me restate. For $p > 2/3$, $3(1-p) < 1$, so:

$$C(1,2) = 3(1-p)T < T \approx C(3,2) \approx C(4,2),$$
$$C(5,2) = T + n_3(2-3p) < T \approx C(3,2) \approx C(4,2).$$

So $C(1,2) < C(5,2) < C(4,2) < C(3,2) < T$ for $p > 2/3$.

Hence the fitness ranking under All-1 invasion for $p > 2/3$ is $f_3 > f_4 > f_5 > f_1$: Grim variants all outperform All-M, with Grim being most resistant and Grim-Forgiver least resistant (though more resistant than All-M). □

Remark 6.2. *Within $\mathcal{F}$ the replicator dynamics has no drift (Remark 6.1), so Proposition 6.4 describes the relative* resistance *to All-*1 *invasion, not an internal ordering within $\mathcal{F}$.*

6.4 Phase Portraits

Because the simplex S_5 is four-dimensional, we represent the dynamics using a *pentagon projection*: each strategy σ_m is placed at the m-th vertex of a regular pentagon, and a population state $\mathbf{x} = (x_1, \ldots, x_5)$ is projected to the 2-D barycentric coordinate $(X, Y) = \sum_m x_m(\cos\theta_m, \sin\theta_m)$ where $\theta_m = \pi/2 - 2\pi(m-1)/5$. The centroid of the pentagon corresponds to the equal-mix state $\mathbf{x} = (1/5, 1/5, 1/5, 1/5, 1/5)$. See Figure 1.

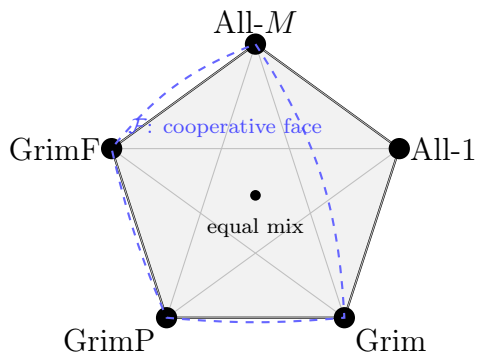
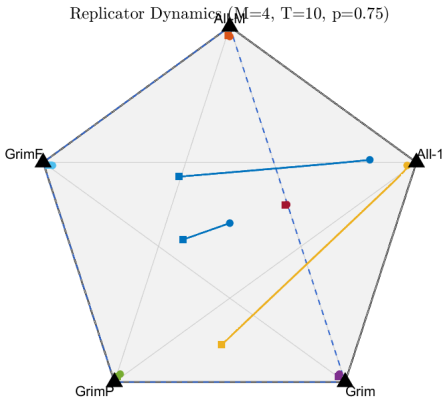
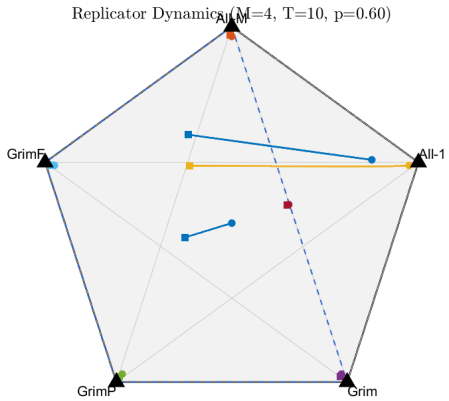


Figure 1: Pentagon representation of the five-strategy simplex S_5. Each vertex is a pure strategy state. The centre corresponds to the equal-frequency population. The dashed region $\mathcal{F}$ (vertices $\sigma_1, \sigma_3, \sigma_4, \sigma_5$) is the cooperative face; all points in $\mathcal{F}$ are rest points of the replicator dynamics. Arrows from MATLAB simulations are overlaid when the `phasePlot5.m` script is run.

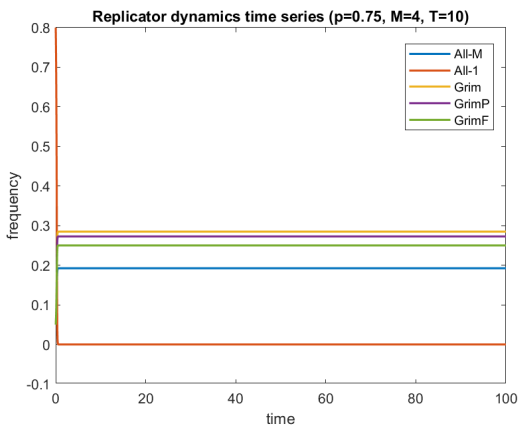


(a) $p = 3/4$: All-1 is stable; orbits from the interior converge to vertex σ_2 or to $\mathcal{F}$.

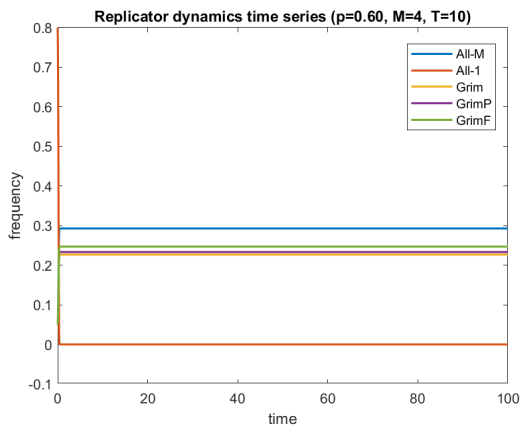


(b) $p = 3/5$: All-1 is unstable; orbits from the interior converge to $\mathcal{F}$ (All-1-free face).

Figure 2: Phase portraits of the replicator dynamics on the pentagon, $M = 4$, $T = 10$. Start points marked with circles; endpoints with squares. Black triangles: pure-strategy vertices.



(a) $p = 3/4$.



(b) $p = 3/5$.

Figure 3: Replicator dynamics time series, $M = 4$, $T = 10$, random initial condition $\mathbf{x}(0) \in S_5$. Line colours: All-M blue, All-1 red, Grim green, GrimP orange, GrimF purple.

6.5 Parameter Sensitivity

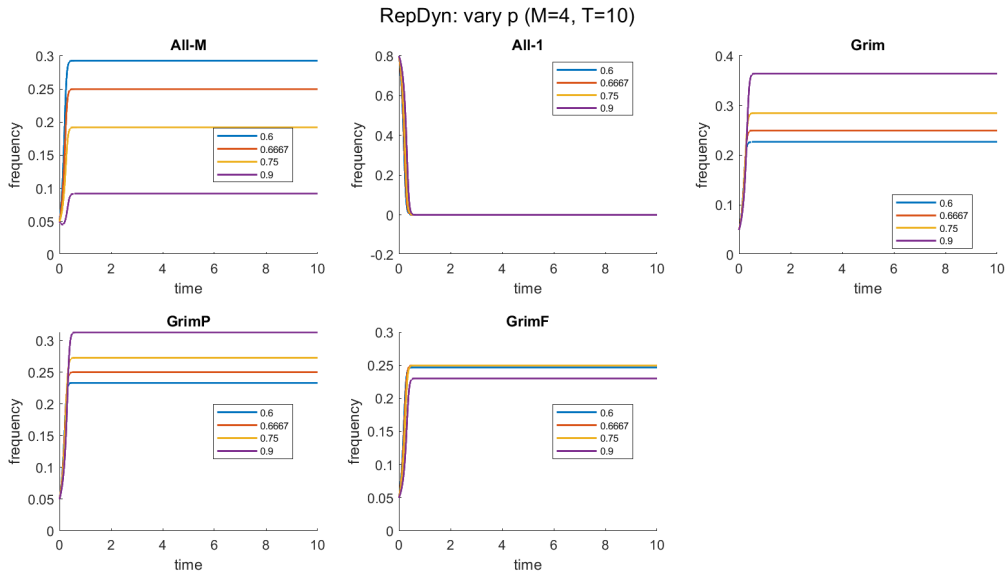


Figure 4: Effect of $p \in \{3/5,\ 2/3,\ 3/4,\ 9/10\}$ on replicator dynamics ($M = 4$, $T = 10$).

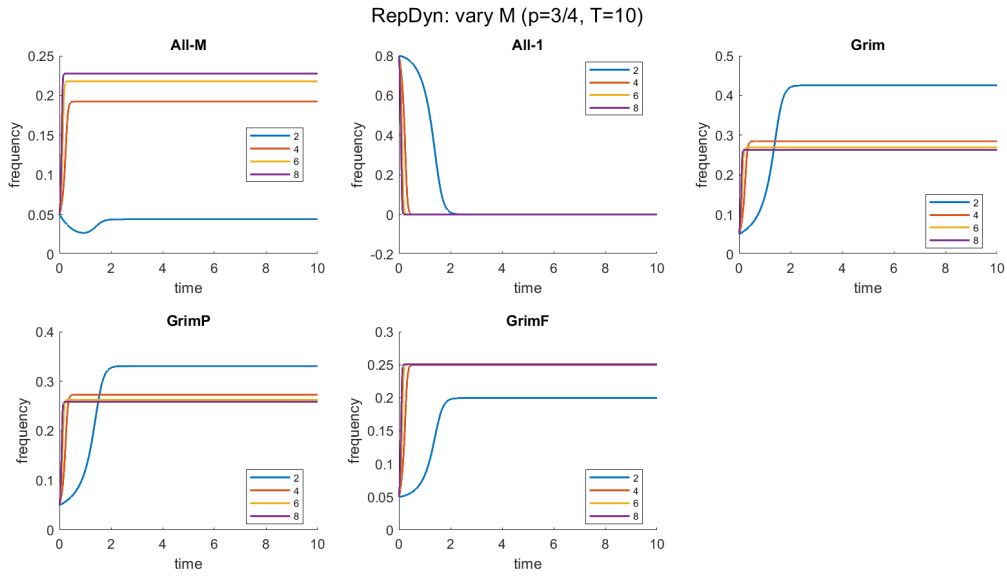


Figure 5: Effect of $M \in \{2, 4, 6, 8\}$ on replicator dynamics ($p = 3/4$, $T = 10$).

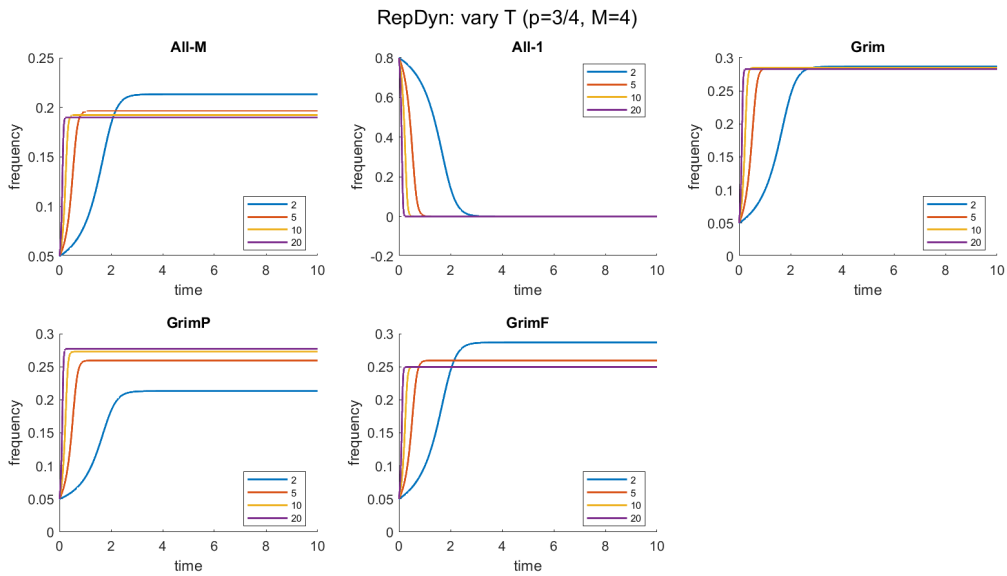


Figure 6: Effect of $T \in \{2, 5, 10, 20\}$ on replicator dynamics ($p = 3/4$, $M = 4$).

7 Evolutionary Stable Strategies

Definition 7.1 (ESS)**.** *A strategy* $\hat{\mathbf{x}} \in S_5$ *is an* evolutionarily stable strategy *(ESS) iff for all* $\mathbf{x} \neq \hat{\mathbf{x}}$*:*

$$\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} \geq \mathbf{x}^\top C\hat{\mathbf{x}}, \tag{7}$$

$$\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} = \mathbf{x}^\top C\hat{\mathbf{x}} \;\Rightarrow\; \hat{\mathbf{x}}^\top C\mathbf{x} > \mathbf{x}^\top C\mathbf{x}. \tag{8}$$

Proposition 7.1 (No Pure ESS from $\mathcal{M}$)**.** *For* $M \geq 4$ *and large* T*, no pure strategy* σ_i *with* $i \in \{1, 3, 4, 5\}$ *is an ESS.*

Proof. Fix $\hat{\mathbf{x}} = \mathbf{e}_i$ for any $i \in \{1, 3, 4, 5\}$. Then $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} = C(i, i) = TM$. For any other $\mathbf{x} = \mathbf{e}_j$ with $j \in \mathcal{M}$, $j \neq i$: $\mathbf{x}^\top C\hat{\mathbf{x}} = C(j, i) = TM = \hat{\mathbf{x}}^\top C\hat{\mathbf{x}}$. So condition (7) holds with equality; we must check (8). But also $\hat{\mathbf{x}}^\top C\mathbf{x} = C(i, j) = TM = \mathbf{x}^\top C\mathbf{x} = C(j, j) = TM$. So (8) fails. Hence $\mathbf{e}_i$ is not an ESS. □

Proposition 7.2 (All-1 is ESS iff $p > \frac{2}{3}$)**.** *For* $M \geq 4$ *and large* T*,* $\mathbf{e}_2 = (0, 1, 0, 0, 0)$ *is an ESS if and only if* $p > \frac{2}{3}$*.*

Proof. $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} = C(2, 2) = T$. For $\mathbf{x} = (a_1, a_2, a_3, a_4, a_5)$:

$$\mathbf{x}^\top C\hat{\mathbf{x}} = \sum_m a_m C(m, 2) = 3(1-p)T\, a_1 + T\, a_2 + (T+2-3p)a_3 + (T+4-6p)a_4 + (T+n_3(2-3p))a_5.$$

For condition (7) to hold we need $T \geq \mathbf{x}^\top C\hat{\mathbf{x}}$, i.e. $\sum_{m\neq 2} a_m(C(m, 2) - T) \leq 0$. Since $C(1, 2) - T = T(2 - 3p)$, $C(3, 2) - T = 2 - 3p$, $C(4, 2) - T = 2(2 - 3p)$, $C(5, 2) - T = n_3(2 - 3p)$: all factors $(2 - 3p)$ are negative iff $p > 2/3$, confirming (7). When $p > 2/3$ the inequality is strict for any $\mathbf{x}$ with $a_m > 0$ for some $m \neq 2$, so (8) is not needed. For $p < 2/3$ at least one factor is positive, violating (7). □

Proposition 7.3 (No ESS in the Cooperative Face)**.** *For* $M \geq 4$ *and large* T*, no mixed strategy in* $\mathcal{F}$ *is an ESS.*

Proof. Let $\hat{\mathbf{x}} \in \mathcal{F}$ (so $x_2 = 0$). $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} = TM$ and for any $\mathbf{x} = \hat{\mathbf{x}} + \delta(\mathbf{e}_j - \hat{x}_j\mathbf{1})$ with $j \in \mathcal{M}$: $\mathbf{x}^\top C\hat{\mathbf{x}} = TM = \hat{\mathbf{x}}^\top C\hat{\mathbf{x}}$ (since all cooperating strategies earn TM against $\hat{\mathbf{x}}$). Condition (7) holds with equality. Then (8) requires $\hat{\mathbf{x}}^\top C\mathbf{x} > TM$, but $\hat{\mathbf{x}}^\top C\mathbf{x} = TM$ (all cooperating). Hence (8) fails for any perturbation within $\mathcal{F}$. □

Corollary 7.1. *The five-strategy ISC has at most one ESS:* $(0, 1, 0, 0, 0)$*, which is ESS iff* $p > \frac{2}{3}$.

8 Markov Chain Analysis

8.1 Setup

The population state is $\mathbf{s} = (s_1, s_2, s_3, s_4, s_5)$ with $\sum_m s_m = N$ and $s_m \in \{0, \ldots, N\}$. The state space is $S_5^N = \{\mathbf{s} \in \mathbb{Z}_{\geq 0}^5 : \sum s_m = N\}$, which has $\binom{N+4}{4}$ elements (e.g. 1001 states for $N = 10$).

The per-player payoff of strategy m at state $\mathbf{s}$ is:

$$q_m(\mathbf{s}) = (s_m - 1)\, C(m, m) + \sum_{n \neq m} s_n\, C(m, n). \tag{9}$$

Definition 8.1 (PPI Protocol)**.** *Under* Pairwise Proportional Imitation *(PPI), a player using strategy* m_1 *switches to* m_2 *with probability*

$$\varrho_{m_1 m_2}(\mathbf{s}) = \frac{x_{m_2}[q_{m_2} - q_{m_1}]_+}{\sum_m x_m [q_m - q_{m_1}]_+},$$

where $[y]_+ = \max(y, 0)$ *and* $x_m = s_m/N$*. The state transition* $\mathbf{s} \to \mathbf{s} - \mathbf{e}_{m_1} + \mathbf{e}_{m_2}$ *has probability* $(s_{m_1}/N)\, \varrho_{m_1 m_2}(\mathbf{s})$.

8.2 Absorbing States

Proposition 8.1 (Absorbing States)**.** *For* $M \geq 4$*,* $N \geq 2$*, and large* T*, the absorbing states of the ISC Markov chain are:*

1. *The pure All-*1 *state* $(0, N, 0, 0, 0)$.
2. *All states on the cooperative face:*

$$\mathcal{A} = \{(s_1, 0, s_3, s_4, s_5) : s_1 + s_3 + s_4 + s_5 = N,\ s_m \geq 0\}. \tag{10}$$

There are $\binom{N+3}{3} + 1$ *absorbing states in total.*

*Proof. Pure All-*1. At $(0, N, 0, 0, 0)$: $q_2(\mathbf{s}) = (N-1)\, C(2, 2)$ and all other q_m are undefined or zero ($s_m = 0$); the only possible switch is to another All-1 player, which is the same strategy. Hence absorbing.

Cooperative face $\mathcal{A}$. At any $\mathbf{s} \in \mathcal{A}$, $s_2 = 0$. For any $m, n \in \{1, 3, 4, 5\}$: $q_m = (s_m - 1)TM + \sum_{n' \neq m} s_{n'} TM = (N-1)TM$ — all strategies earn the same payoff. Hence $[q_n - q_m]_+ = 0$ for all $n \neq m$; no player ever switches. Any attempt to switch *toward* All-1 is also impossible since $s_2 = 0$ and a non-existent strategy cannot be imitated. Hence $\mathcal{A}$ is absorbing.

Counting. $|\mathcal{A}| = \binom{N+3}{3}$ (four-part compositions of N) plus the one extra state $(0, N, 0, 0, 0)$. □

Remark 8.1. *This is a profound generalisation of the original paper [2]. In the three-strategy version with All-M, the $s_2 = 0$ line (a 1-D face) was absorbing. With five strategies the entire $s_2 = 0$ four-dimensional face $\mathcal{A}$ is absorbing. The key is that all four cooperating strategies earn identical payoffs TM against each other.*

8.3 Transient States and Absorption

Proposition 8.2 (Interior States are Transient)**.** *For $M \geq 4$, $N \geq 4$, and large T, any state $\mathbf{s}$ with $s_2 \geq 1$ and at least one $s_m > 0$ for $m \in \{1, 3, 4, 5\}$ is transient.*

Proof. At such a state, at least one player uses All-1. We show that transitions strictly reducing s_2 have positive probability.

The payoff difference between strategy $m \in \{1, 3, 4, 5\}$ and All-1:

$$q_m - q_2 = (s_m - 1) \cdot TM + s_2\, C(m, 2) + \sum_{n \notin \{2,m\}} s_n TM - (s_2 - 1)\, T - \sum_{m' \neq 2} s_{m'}\, C(2, m').$$

To leading order in T:

$$q_m - q_2 \approx T\Big[(N - 1 - s_2)M + s_2 c_m - (s_2 - 1) - (N - 1 - s_2) \cdot 3p\Big],$$

where $c_m = C(m, 2)/T$ (from row m of (4)). For $m = 3$: $c_3 \approx 1$.

$$q_3 - q_2 \approx T(N - 1 - s_2)(M - 3p) + T\, s_2(1 - 1) = T(N - 1 - s_2)(M - 3p) > 0$$

since $M \geq 4 > 3p$ and $s_2 < N - 1$. Hence a Grim player can always switch *to* All-1 ... wait that is the wrong direction.

Let me redo: we want to show $q_m > q_2$ so that All-1 players switch to m.

$$q_m - q_2 \approx T\big[(N - 1 - s_2)(M - 3p) + s_2(c_m - 1)\big].$$

For $m = 3$, $c_3 \approx 1$: $q_3 - q_2 \approx T(N - 1 - s_2)(M - 3p) > 0$. For $m = 1$, $c_1 = 3(1 - p)$: $q_1 - q_2 \approx T[(N - 1 - s_2)(M - 3p) + s_2(3(1 - p) - 1)]$. The first term dominates for $s_2 < N - 1$. So $q_m > q_2$ for all $m \in \mathcal{M}$ when $s_2 < N - 1$.

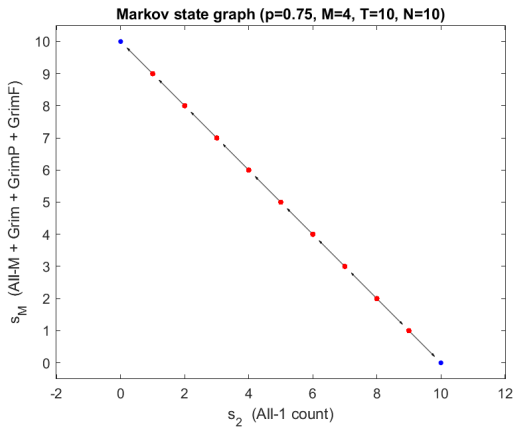
Hence $\varrho_{2m} > 0$: All-1 players imitate cooperating strategies. The transition $\mathbf{s} \to \mathbf{s} - \mathbf{e}_2 + \mathbf{e}_m$ has positive probability, reducing s_2. Since s_2 is non-negative, by induction the chain is eventually absorbed in $\mathcal{A} \cup \{(0, N, 0, 0, 0)\}$.

Whether absorption is into $(0, N, 0, 0, 0)$ or $\mathcal{A}$ depends on whether All-1 players switch to cooperating strategies before they have colonised the full population. □

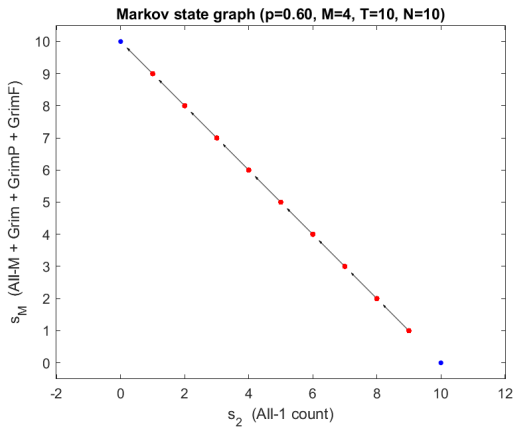
Proposition 8.3 (Absorption into All-1 Equilibrium)**.** *For $M \geq 4$, large T, and $p > \frac{2}{3}$, the state $(0, N - 1, 1)$ with exactly one cooperator (strategy $m \in \mathcal{M}$) is absorbed into $(0, N, 0, 0, 0)$ almost surely.*

Proof. At $(0, N - 1, 1)$ ($s_2 = N - 1$, $s_m = 1$ for some $m \in \mathcal{M}$): $q_m = 0 \cdot TM + (N - 1) \cdot C(m, 2)$ and $q_2 = (N - 2)T + C(2, m)$. For large T: $q_m \approx (N - 1) \cdot T \cdot c_m$ and $q_2 \approx (N - 2)T + T \approx (N - 1)T$. So $q_m - q_2 \approx T(N - 1)(c_m - 1)$. For $m = 3$: $c_3 \approx 1$, so $q_3 - q_2 \approx 0$; the $O(1)$ term determines the sign: $q_3 - q_2 = (N - 1)(2 - 3p)$. For $p > 2/3$: $2 - 3p < 0$, so $q_3 < q_2$. The unique cooperator is less fit than All-1 and switches away (with positive probability), leading to absorption in $(0, N, 0, 0, 0)$. □

8.4 State Transition Diagrams

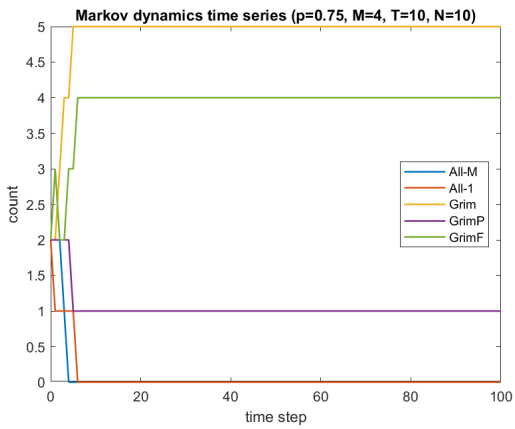


(a) $p = 3/4$: drift toward All-1 vertex and absorbing face $\mathcal{A}$.

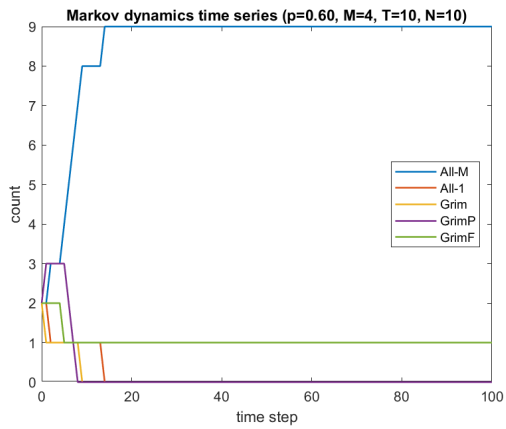


(b) $p = 3/5$: drift entirely toward absorbing face $\mathcal{A}$.

Figure 7: Markov state transition diagrams projected onto a 2-D subspace (s_2 vs. $s_{\mathcal{M}} = s_1 + s_3 + s_4 + s_5$) for $N = 10$, $M = 4$, $T = 10$. Blue: absorbing states. Red: transient.



(a) $p = 3/4$.



(b) $p = 3/5$.

Figure 8: Markov dynamics time series, $M = 4$, $T = 10$, $N = 10$, 10^4 steps, initial state $\mathbf{s}_0 = (2, 2, 2, 2, 2)$.

8.5 Parameter Sensitivity

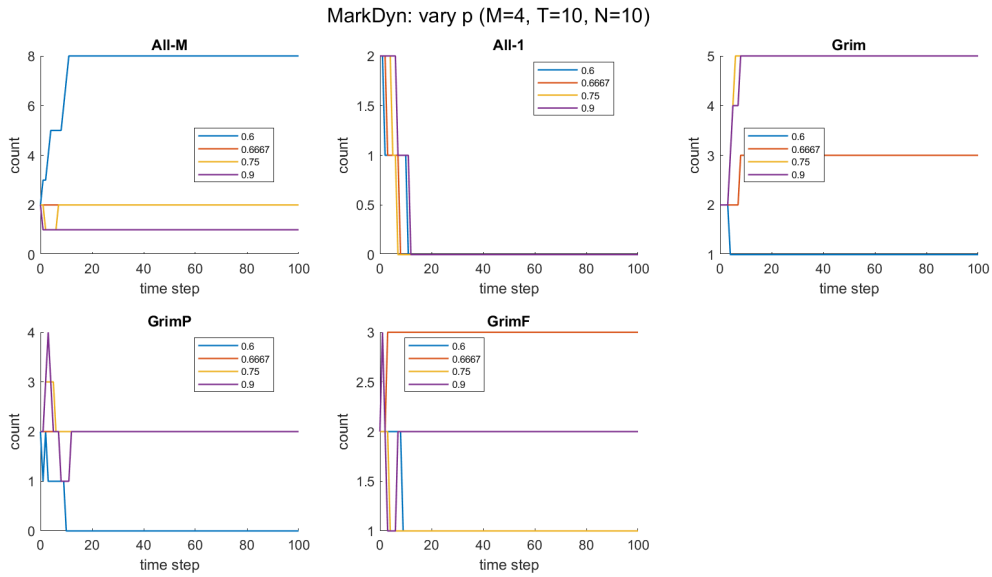


Figure 9: Effect of $p \in \{3/5, 2/3, 3/4, 9/10\}$ on Markov dynamics ($M = 4$, $T = 10$, $N = 10$).

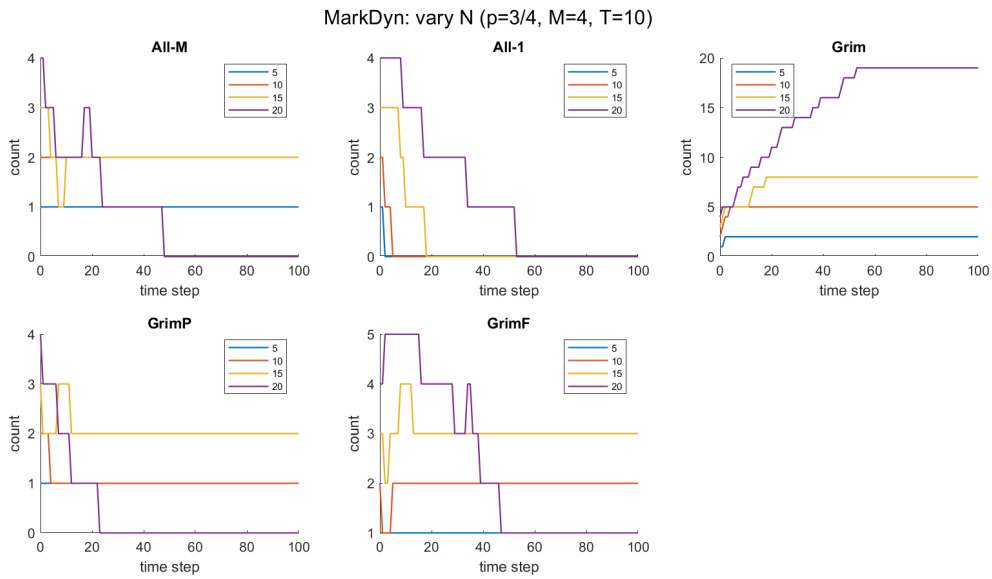


Figure 10: Effect of $N \in \{5, 10, 15, 20\}$ on Markov dynamics ($p = 3/4$, $M = 4$, $T = 10$).

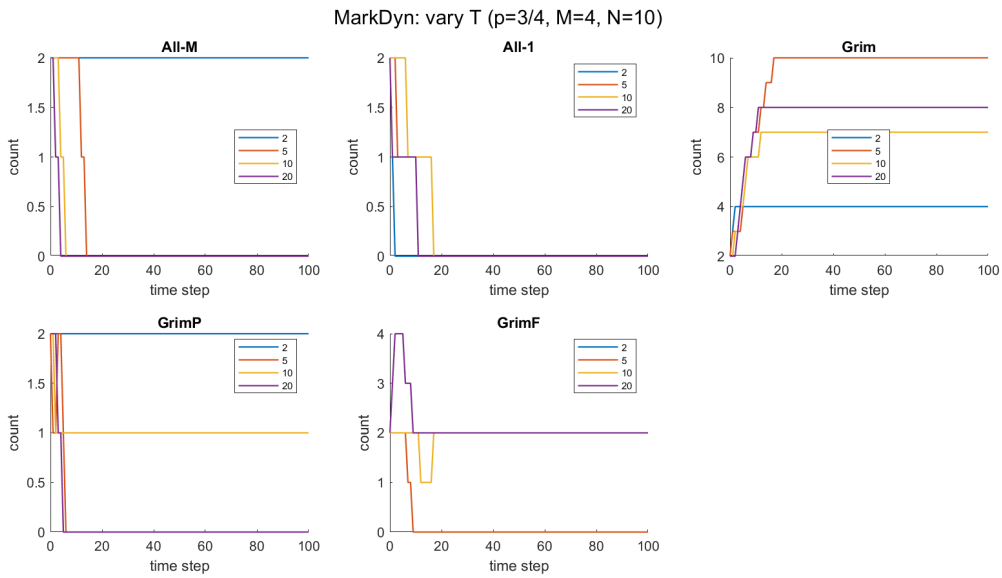


Figure 11: Effect of $T \in \{2, 5, 10, 20\}$ on Markov dynamics ($p = 3/4$, $M = 4$, $N = 10$).

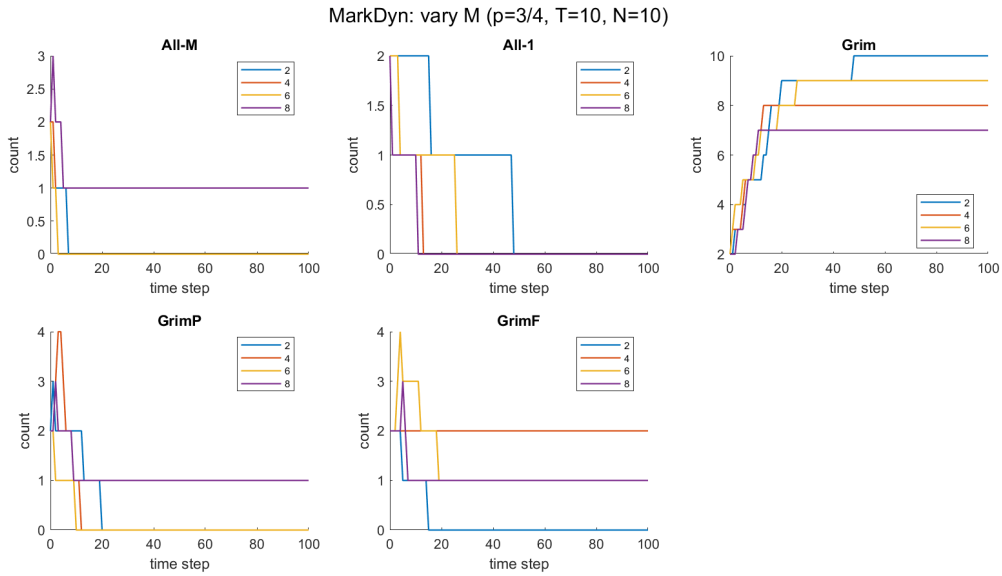


Figure 12: Effect of $M \in \{2, 4, 6, 8\}$ on Markov dynamics ($p = 3/4$, $T = 10$, $N = 10$).

9 Connection to Learning Theory

Both evolutionary game theory (EGT) and individual learning models describe adaptive behaviour through the same formal dynamics but under different interpretations [4, 6].

In EGT, $x_m(t)$ denotes the population share of strategy m, and the replicator dynamic (5) describes how more successful strategies grow.

In a learning model, individual agents update mixed strategies based on accumulated payoff experience. Under *reinforcement learning*, the propensity q_m to play strategy m evolves as $q_m(t+1) = q_m(t) + \alpha\pi_m(t)$ with $x_m(t) = q_m(t)/\sum_n q_n(t)$. As shown in [4], this rule converges in large populations to the replicator dynamic.

In the five-strategy ISC context this duality has the following interpretation:

- **Evolutionary.** The four cooperating strategies are *evolutionarily neutral* relative to each other (equal fitness TM on $\mathcal{F}$). Natural selection acts only on the All-1/cooperator axis, with threshold $p^* = 2/3$.

- **Learning.** A rational agent learning from experience will reinforce any cooperating strategy over All-1 when $p < 2/3$, because cooperation yields higher average payoff (TM vs T). Among cooperating strategies, reinforcement is indifferent (equal payoffs), so the agent's internal mix over {All-M, Grim, GrimP, GrimF} is determined by initial conditions and random variation.

- **Finite-population.** PPI corresponds to *social learning*: agents imitate better-performing neighbours. The absorbing face $\mathcal{A}$ captures the long-run state where social learning has eliminated All-1 from the population.

10 Conclusion

We have studied the Iterated Symmetric Centipede game with five strategies: All-M, All-1, Grim, Grim-Patient, and Grim-Forgiver. The main results are:

1. **Payoff matrix.** The 5×5 ISC payoff matrix C (Proposition 4.1) has a highly degenerate structure: all four cooperating strategies earn TM against each other, and differ only in their interaction with All-1. Grim-Forgiver is the most exploitable: $C(2,5) > C(2,4) > C(2,3) > C(2,2)$.

2. **Nash equilibria.** The ISC has a continuum of Nash equilibria: any mixture over {All-M, Grim, GrimP, GrimF} is a NE. Additionally, (σ_2, σ_2) is a NE iff $p \geq 2/3$ (Propositions 5.1 and 5.2).

3. **Replicator dynamics.** The cooperative face $\mathcal{F} = \{x_2 = 0\}$ is a rest-point set with no internal drift. The All-1 equilibrium is stable iff $p > 2/3$ (Proposition 6.3).

4. **ESS.** No pure cooperating strategy is an ESS (Proposition 7.1). All-1 is an ESS iff $p > 2/3$ (Proposition 7.2). No mixed strategy in $\mathcal{F}$ is an ESS (Proposition 7.3).

5. **Markov dynamics.** The absorbing states form the $\binom{N+3}{3}$-element cooperative face $\mathcal{A} = \{s_2 = 0\}$ plus the pure All-1 state (Proposition 8.1). All states with $s_2 \geq 1$ and at least one cooperator are transient.

6. **Phase portraits.** A pentagon projection onto 2-D is introduced for visualisation; orbits converge either to the pentagon face (cooperative region) or to the All-1 vertex.

7. **Comparing Grim variants.** Grim is most resistant to All-1 exploitation; Grim-Forgiver is least resistant but also least aggressive (allows partial recovery of cooperation after punishment).

Future work includes: (a) adding Tit-for-Tat and Generous-Tit-for-Tat strategies; (b) studying the effect of noise/mutation on the neutrally stable cooperative face; (c) analysing asymmetric populations.

References

[1] R. W. Rosenthal (1981). "Games of perfect information, predatory pricing and the chain-store paradox." *Journal of Economic Theory*, 25, 92–100.

[2] I. Lazaridis and A. Kehagias (2026). *Iterated Symmetric Centipede: Learning and Evolutionary Games*. Preprint, March 2026.

[3] R. Axelrod and W. D. Hamilton (1981). "The evolution of cooperation." *Science*, 211, 1390–1396.

[4] T. Börgers and R. Sarin (1997). "Learning through reinforcement and replicator dynamics." *Journal of Economic Theory*, 77, 1–14.

[5] J. Hofbauer and K. Sigmund (1998). *Evolutionary Games and Population Dynamics*. Cambridge University Press.

[6] W. H. Sandholm (2010). *Population Games and Evolutionary Dynamics*. MIT Press.

[7] J. G. Kemeny and J. L. Snell (1969). *Finite Markov Chains*. Van Nostrand, Princeton.

[8] R. Cressman (2003). *Evolutionary Dynamics and Extensive Form Games*. MIT Press.

[9] E. M. Krockow, A. M. Colman and B. D. Pulford (2016). "Cooperation in repeated interactions: A systematic review of Centipede game experiments, 1992–2016." *European Review of Social Psychology*, 27, 231–282.

[10] W. H. Sandholm, S. S. Izquierdo and L. R. Izquierdo (2019). "Best experienced payoff dynamics and cooperation in the centipede game." *Theoretical Economics*, 14, 1347–1385.

A Additional Strategies

The framework is easily extended. Further strategies include:

1. **Always-m:** always play ρ_m for fixed $m \in \{1, \ldots, M\}$.
2. **Tit-for-Tat:** begin with ρ_M; in each subsequent round play whatever OSC action the opponent used in the previous round.
3. **Generous-Tit-for-Tat:** as Tit-for-Tat but forgive defection with probability $q = (1-p)/(pM - (1-p))$.

B Two-Population Replicator Dynamics

The one-population replicator dynamics studied in Section 6 necessarily yields symmetric NE. Non-symmetric NE can be recovered with two-population dynamics. Let $\mathbf{x}(t)$ and $\mathbf{y}(t)$ be the strategy frequency vectors of Population 1 and Population 2 respectively. The two-population replicator dynamics is:

$$\dot{x}_m = x_m \left[(C\mathbf{y})_m - \mathbf{x}^\top C\mathbf{y} \right], \tag{11}$$

$$\dot{y}_m = y_m \left[(C^\top \mathbf{x})_m - \mathbf{y}^\top C^\top \mathbf{x} \right]. \tag{12}$$

A stability analysis of (11)–(12) for the five-strategy ISC is left for future work.

C Pentagon Barycentric Coordinates

The pentagon projection maps a frequency vector $\mathbf{x} = (x_1, \ldots, x_5) \in S_5$ to:

$$(X, Y) = \sum_{m=1}^{5} x_m \left(\cos\theta_m,\, \sin\theta_m\right), \quad \theta_m = \frac{\pi}{2} - \frac{2\pi(m-1)}{5}.$$

Key properties:

- Pure state $\mathbf{e}_m$ maps to vertex $(\cos\theta_m, \sin\theta_m)$.
- Equal mix $(1/5, \ldots, 1/5)$ maps to the centroid $(0, 0)$.
- The cooperative face $\mathcal{F} = \{x_2 = 0\}$ maps to the convex hull of $\{\theta_1, \theta_3, \theta_4, \theta_5\}$, which is the pentagon with the All-1 vertex removed.
- The map is many-to-one: different $\mathbf{x}$ may have the same image. Trajectories crossing in the projected space do not necessarily cross in the simplex.

The MATLAB function `pentaBary.m` implements this projection.